

## 例题1:

石油在长度 $L=50\text{m}$ 、直径 $d=100\text{mm}$ 的管道内流动，管轴对于水平的倾斜度为 $30^\circ$ ， $\mu=0.285\text{Ns/m}^2$ ， $\rho=950\text{kg/m}^3$ ， $\text{Re}_{cr}=2000$

试确定

- (1) 为保持层流状态的最大允许流量；
- (2) 相应的进出口压差。

解：(1)若流动管内层流 (即Hagen-Poiseuille流)

$$\text{由 } \text{Re}_{cr} = \frac{\rho \bar{V} d}{\mu} = 2000$$

$$\text{得到 } \bar{V} = 6 \text{ m/s}, \quad Q = \bar{V} \frac{\pi d^2}{4} = 0.047 \text{ m}^3/\text{s}$$

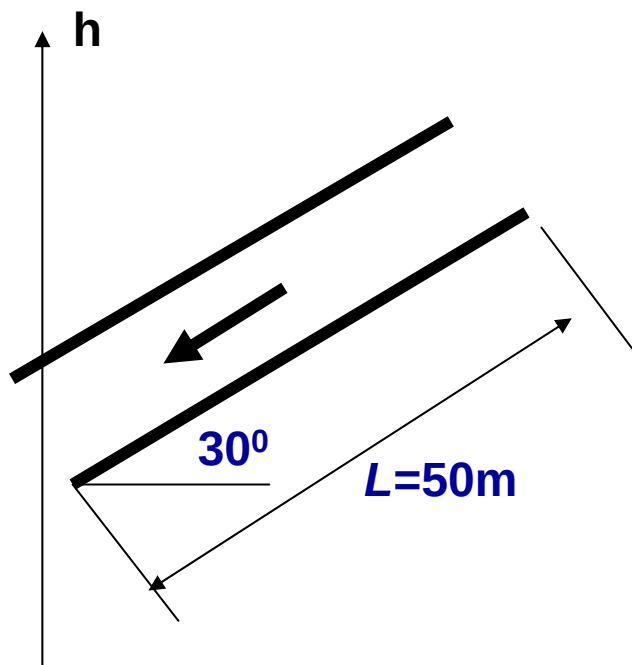
(2) 相应的进出口压差

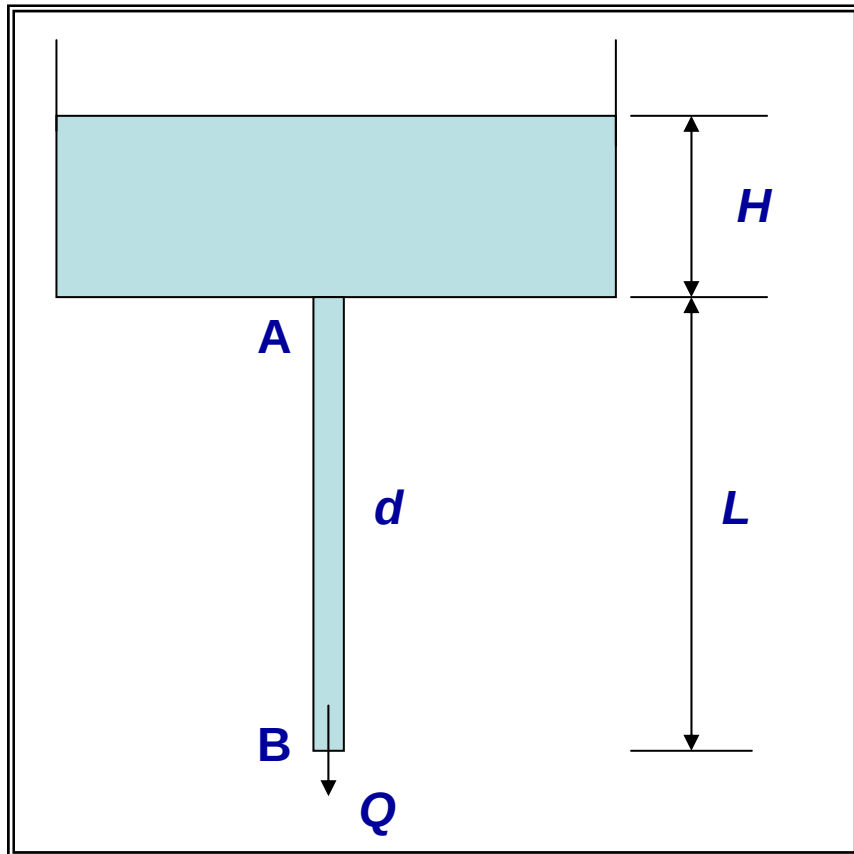
$$\Delta p^* = \lambda \frac{1}{2} \rho \bar{V}^2 \frac{L}{d} = \frac{64}{\text{Re}_{cr}} \frac{1}{2} \rho \bar{V}^2 \frac{L}{d} = 273600 \text{ N/m}^2$$

$$\Delta p^* = \Delta p + \rho g \Delta h$$

$$\Delta h = L \sin \alpha$$

$$\Delta p = 273600 - \rho g \Delta h = 40850 \text{ N/m}^2$$





## 例题2:

毛细管粘度计 (测 $\mu$ )

AB内近似看成定常充分发展层流管流

$$Q = -\frac{\pi d^4}{128\mu} \frac{dp^*}{dx}$$

$$-\frac{dp^*}{dx} = -\frac{p_B + \rho g h_B - (p_A + \rho g h_A)}{L}$$

(做了近似)

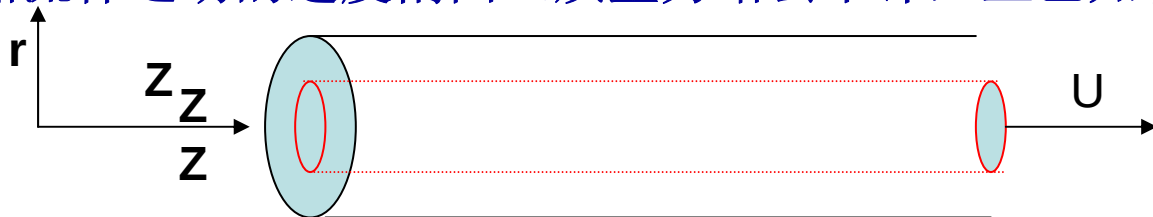
$$= \frac{p_A - p_B + \rho g(h_A - h_B)}{L}$$

$$= \frac{\rho g H + \rho g L}{L}$$

$$\therefore \mu = \frac{\pi d^4}{128Q} \frac{\rho g(H + L)}{L}$$

近似: 无限长, 至少 $L/d > 100$ , 入口效应可不计了  
 定常,  $H$  应恒定, 实际选取一段时间内平均高度近似  
 层流,  $d$  是毫米量级, 流速很慢,  $Re < 2000$   
 圆管, 细选毛细管, 横截面一定  
 理论公式, 如何在实际中应用

**例题3:** 无限长的半径为 $R_1$ 的圆柱，以常速度 $U$ 沿轴线方向在无限长的半径为 $R_2$ 的同轴圆柱面内运动，试确定由此引起的两柱面间的粘性不可压缩流体运动的速度剖面（质量力略去不计，且已知轴向压力梯度为零）



**解:** 采用柱坐标系 $(r, \varepsilon, z)$  轴与轴线重合

由题意知:  $V_r = V_\varepsilon = 0$   $V_z = V$

$$\frac{\partial}{\partial t} = 0 \quad \frac{\partial}{\partial \varepsilon} = 0 \quad \frac{\partial p}{\partial z} = 0$$

由连续方程知:  $\frac{\partial V}{\partial z} = 0 \Rightarrow V = V(r)$

由径向运动方程知:  $\frac{\partial p}{\partial r} = 0$  周向运动方程自动满足

由轴向运动方程知:  $\frac{d}{dr} \left( r \frac{dV}{dr} \right) = 0$

积分两次得:  $V = c_1 \ln r + c_2$

边界条件:  $V|_{r=R_1} = U$   $V|_{r=R_2} = 0$   $\longrightarrow$

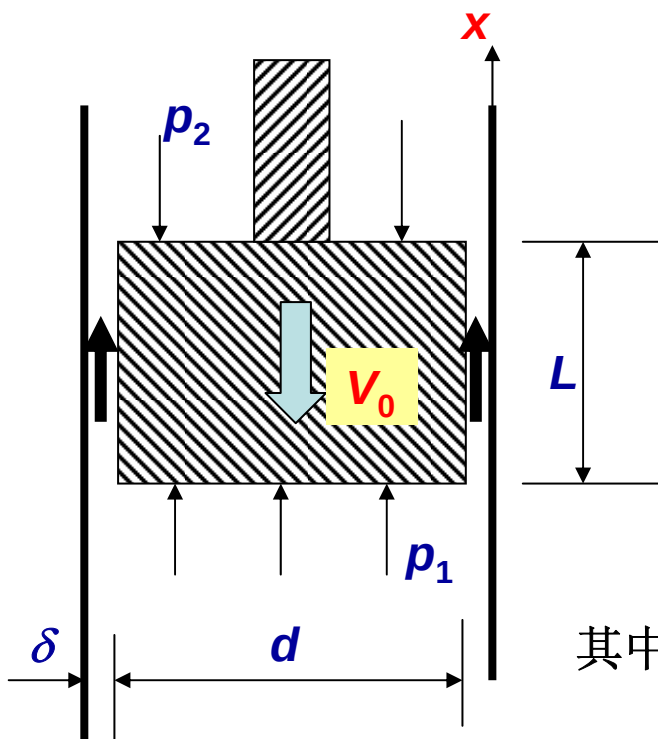
$$c_1 = -\frac{U}{\ln\left(\frac{R_2}{R_1}\right)}$$

$$c_2 = \frac{U \ln R_2}{\ln\left(\frac{R_2}{R_1}\right)}$$

所以:  $V = U \frac{\ln\left(\frac{r}{R_2}\right)}{\ln\left(\frac{R_1}{R_2}\right)}$

**例题4:** 充满油的油缸内有长 $L$ 活塞向下以 $V_0$ 等速运动，油通过间隙 $\delta$ 向上流动， $\delta \ll d$ ，流动是层流，已知油粘度系数 $\mu$ ，密度 $\rho$ ，活塞上下压差 $p_2$ 、 $p_1$ ，活塞直径 $d$ ，间隙 $\delta$ ，忽略端部效应，求 $V_0$ 。

解：环形通道， $\delta \ll d$ ，缝隙内看成无限大两平板间的Couette流动



缝隙内流动方向为 $x$ 向， $U = -V_0$ （要求的量）

$$Q = V_0 \frac{\pi}{4} d^2$$

[couette  $Q = b \int_0^H u dy = b \left( -\frac{1}{12\mu} \frac{dp^*}{dx} H^3 + \frac{UH}{2} \right)$ ]

$$\delta \ll d \quad b = \pi d, \quad H = \delta$$

$$Q = \pi d \left( -\frac{1}{12\mu} \frac{dp^*}{dx} \delta^3 - \frac{V_0 \delta}{2} \right) = V_0 \frac{\pi}{4} d^2 \quad \text{😊}$$

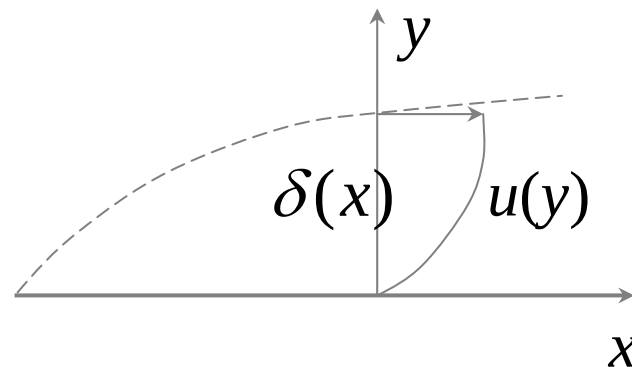
其中  $\frac{dp^*}{dx} = \frac{\Delta(p + \rho gh)}{L} = \frac{(p_2 + \rho gL) - p_1}{L} = \frac{p_2 - p_1}{L} + \rho g$

由😊得到  $V_0 \left( \frac{\delta}{2} + \frac{\pi d^2}{4} \right) = -\frac{\delta^3}{12\mu L} (p_2 - p_1 + \rho gL)$

$$V_0 = \frac{\delta^3}{3\mu(2\delta + \pi d^2)} \frac{(p_1 - p_2 - \rho gL)}{L} \approx \frac{\delta^3}{3\mu\pi d^2 L} (p_1 - p_2 - \rho gL)$$

### 例题5

$$\frac{u}{U} = a + b \frac{y}{\delta} + c \left( \frac{y}{\delta} \right)^2$$



$$\begin{aligned} BC: \quad y=0 \quad u=0 \\ y=\delta \quad u \approx U \quad \frac{\partial u}{\partial y} \approx 0 \end{aligned}$$

$$a=0, b=\frac{2U}{\delta}, c=-\frac{U}{\delta^2}$$

$$\frac{u}{U} = \frac{2y}{\delta} - \frac{y^2}{\delta^2}$$

$$0 \leq y \leq \delta(x)$$

$$\delta_2 = \int_0^\delta \frac{u}{U} \left(1 - \frac{u}{U}\right) dy \qquad \frac{u}{U} = \frac{2y}{\delta} - \frac{y^2}{\delta^2}$$

$$\delta_2 = \int_0^\delta \left(\frac{2y}{\delta} - \frac{y^2}{\delta^2}\right) \left(1 - \frac{2y}{\delta} + \frac{y^2}{\delta^2}\right) dy \qquad \text{let } \eta = \frac{y}{\delta}$$

$$\delta_2 = \delta \int_0^1 (2\eta - \eta^2)(1 - 2\eta + \eta^2) d\eta$$

$$= \delta \int_0^1 (-\eta^4 + 4\eta^3 - 5\eta^2 + 2\eta) d\eta = \frac{2}{15} \delta \qquad \boxed{\delta_2 = \frac{2}{15} \delta}$$

$$\tau_w = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} = \frac{\mu U}{\delta} \left. \frac{\partial (u/U)}{\partial (y/\delta)} \right|_{y=0}$$

$$\boxed{\tau_w(x) = \frac{2\mu U}{\delta}}$$

**例6：** 有一块  $1.5\text{ m} \times 4.5\text{ m}$  的矩形薄板在空气中以  $3\text{ m/s}$  速度沿板面方向拖动，已知空气运动粘性系数为  $\nu = 1.5 \times 10^{-5}\text{ m}^2/\text{s}$ ，密度为  $\rho = 1.2\text{ kg/m}^3$ 。试求薄板沿短边方向和长边方向运动时，各自的摩擦阻力。

解：（1）求沿短边方向运动的阻力。

$$\text{Re}_L = \frac{UL}{\nu} = \frac{3 \times 1.5}{1.5 \times 10^{-5}} = 3 \times 10^5$$

$$\text{转捩点: } x_k = \text{Re}_{x_k} \frac{\nu}{U_0} = 5 \times 10^5 \times \frac{1.5 \times 10^{-5}}{3} = 2.5\text{ m}$$

因为  $L_1 = 1.5\text{ m} < 2.5\text{ m}$ ，所以全板为层流边界层

$$C_{f_1} = \frac{1.33}{\text{Re}_{L_1}^{0.5}} = \frac{1.33}{\sqrt{\frac{3 \times 1.5}{1.5 \times 10^{-5}}}} = 0.00242$$

$$D_{f_1} = 2C_{f_1} bL \frac{\rho}{2} U_0^2 = 2 \times 0.00242 \times 4.5 \times 1.5 \times \frac{1.2}{2} 3^2 = 0.176\text{ N}$$

(2) 求沿长边运动时的阻力

$L_2 > x_k$  , 为混合附面层

$$\text{Re}_L = \frac{U_0 L_2}{\nu} = \frac{3 \times 4.5}{1.5 \times 10^{-5}} = 9 \times 10^5 < 10^6$$

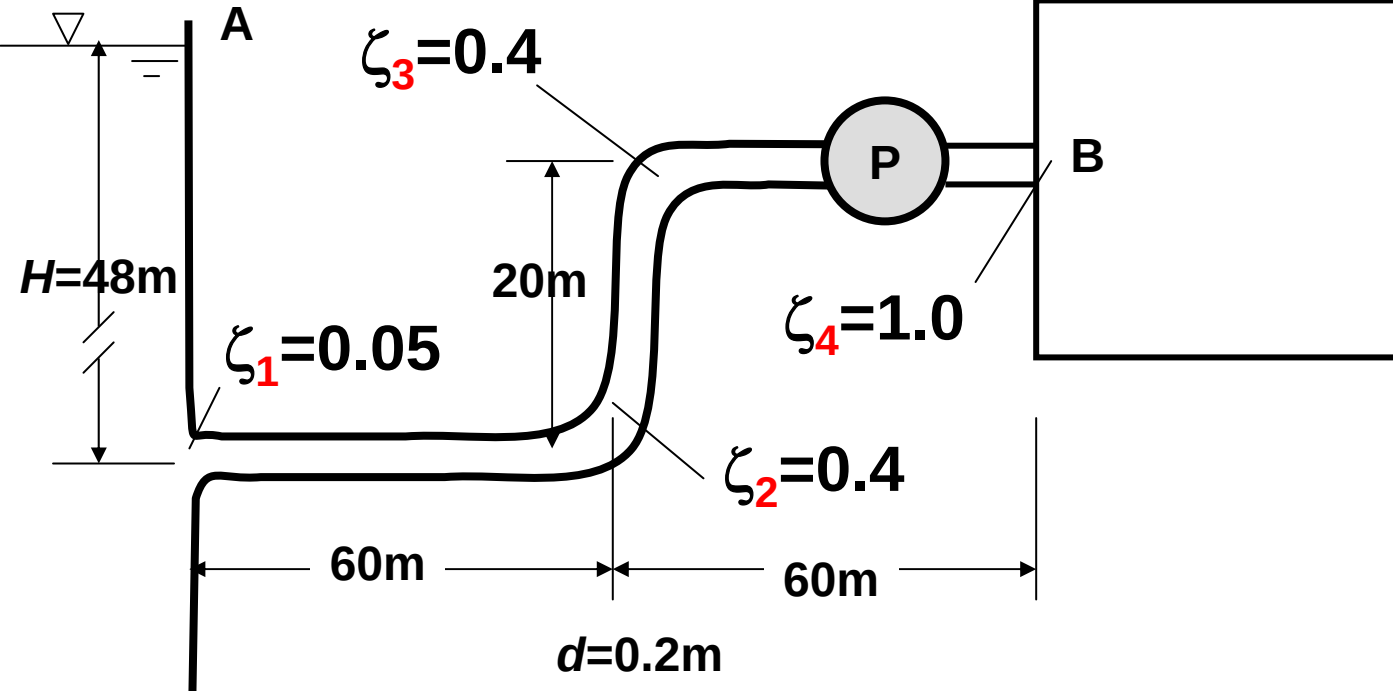
$$\begin{aligned} C_{f2} &= \frac{0.074}{\text{Re}_L^{0.2}} - \frac{A}{\text{Re}_L} \\ &= \frac{0.074}{(9 \times 10^5)^{0.2}} - \frac{1700}{9 \times 10^5} = 0.0288 \end{aligned}$$

$$D_{f2} = 2C_{f2} b L_2 \frac{\rho}{2} U_0^2 = 2 \times 0.00288 \times 1.5 \times 4.5 \frac{1.2}{2} 3^2 = 0.21 \text{ N}$$

比较 (1) , (2) 结果, 一块平板混合边界层产生的阻力大于层流边界层的阻力。



# 例题7



已知,  $\mu = 1.14 \times 10^{-3} \text{m}^2/\text{s}$ ,  $\rho = 1000 \text{kg}/\text{m}^3$ , 粗糙度  $h/d = 0.00023$ ,  $N_{\text{pump}} = 20 \text{kW}$ ,  $Q = 0.14 \text{m}^3/\text{s}$   
求: B处的压力 (表压)

$$p_A + \frac{1}{2} \rho V_A^2 + \rho g h_A + \Delta p_{\text{pump}} = p_B + \frac{1}{2} \rho V_B^2 + \rho g h_B + \Delta p_f + \sum \Delta p_\zeta$$

$$\Delta p_{\text{pump}} = \frac{N_{\text{pump}}}{Q} \quad V = V_B = \frac{Q}{\frac{\pi}{4} d^2} \quad Re = \frac{\rho V d}{\mu} = 7.81 \times 10^5 \quad \lambda = 0.0152$$

$$\Delta p_f = \sum_{i=1}^3 \lambda \frac{L_i}{d_i} \frac{\rho V_i^2}{2} \quad \sum \Delta p_\zeta = (\zeta_1 + \zeta_2 + \zeta_3 + \zeta_4) \frac{\rho V^2}{2} \quad p_B - p_a = 283 \text{kPa}$$